Signals and Systems

Grade Test: Quiz 5

Test Information

QUESTION 1: CALCULATED FORMULA

5 **out of 5 points**

Consider the signals x(t), y(t) and z(t) whose regions of convergence are given by ROC1={σ>7}, ROC2={σ>11} and ROC3={σ>22}, respectively.

Let w(t)=x(t)+y(t)+z(t). If the region of convergence of w(t) is given by ROC={σ>A}, what is the value of the constant A?

Given Answer: 22

Correct Answer: 22

QUESTION 2: MULTIPLE CHOICE | 10 | **out of 10 points**

What is the inverse Laplace transform of the function $X(s) = \frac{s-2}{(s+1)^2(s+2)}$ (for positive times)?

Given Answer: 4*exp(-t) - 4*exp(-2*t) - 3*t*exp(-t)

Correct Answer: 4*exp(-t) - 4*exp(-2*t) - 3*t*exp(-t)

QUESTION 3: TRUE/FALSE | 10 | **out of 10 points**

Consider a LTI system whose input x(t) and output y(t) are related by the differential equation: y''(t)+9y'(t)+20y(t)=5x''(t)+4x(t)-x(t). This system is stable.

Given Answer: True

Correct Answer: True

QUESTION 4: MULTIPLE CHOICE | 10 | **out of 10 points**

What is the Laplace transform of the function $x(t) = e^{-2t}u(t-4)$?

Given Answer: (exp(-4*s)*exp(-8))/(s + 2)

Correct Answer: (exp(-4*s)*exp(-8))/(s + 2)

QUESTION 5: TRUE/FALSE | 5 | **out of 5 points**

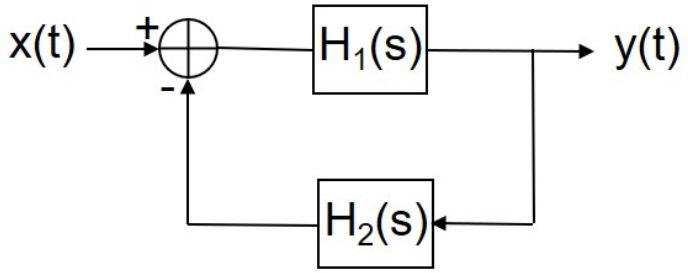
A system whose impulse response is given by $h(t) = te^{-2t}u(t)$ is stable.

Given Answer: True

Correct Answer: True

QUESTION 6: MULTIPLE CHOICE 10 **out of 10 points**

Consider the system shown below where $H_1(s)=a$ and $H_2(s)=1/(s-2)$.
Above which value of the real number a is the system stable?



Given Answer: 2

Correct Answer: 2

QUESTION 7: CALCULATED FORMULA 10 **out of 10 points**

Consider a LTI system whose transfer function can be written as
H(s)=N(s)/D(s) where:

- N(s)=4s+2.
- D(s)=$7s^2$+5s+1.

Find the final value of the step response.

Given Answer: 2

Correct Answer: 2.0

QUESTION 8: TRUE/FALSE 5 **out of 5 points**

A system whose transfer function is given by
$H(s) = \frac{1}{(s-1)(s+2)}$ is stable.

Given Answer: False

Correct Answer: False

QUESTION 9: CALCULATED FORMULA 5 **out of 5 points**

Consider a signal whose Laplace transform is given by $X(s) = \frac{1}{s^2 + 2as + b}$. If x(t) can be written as $Ae^{-Bt}\sin(Ct)u(t)$, what is the value of the constant C^2 if a=6 and b=95? (**attention:** write down the value of $\mathbf{C^2}$ not C).

Given Answer: 59

Correct Answer: 59

QUESTION 10: CALCULATED FORMULA 5 **out of 5 points**

Consider a signal whose Laplace transform is given by $X(s) = \frac{1}{s^2 + 2as + b}$. If x(t) can be written as $Ae^{-Bt}\sin(Ct)u(t)$, what is the value of the constant B if a=3 and b=70?

Given Answer: 3

Correct Answer: 3

FEEDBACK AND NOTES FOR ATTEMPT

Feedback to Learner

Paragraph Arial 3 (12pt)

Mashups

Path: p Words:0

Grading Notes

Paragraph Arial 3 (12pt)

Mashups

Path: p Words:0